HDD Companies Analysis

Seagate, Western Digital, Toshiba and the HDD supply chain — financials, technology, growth potential and downturn survivability
Report 2 of 3 · Prepared 28 October 2026

3
HDD makers left
(Seagate, WD, Toshiba)

44/44/11%
Revenue share 2026
WD / Seagate / Toshiba

Net cash
Seagate (by end-2026) and WD
both investment-grade or near

Survive
All three survive a 2023-style downturn; layoffs likely

Bottom line

Seagate and Western Digital are both strong, long-lived businesses and would stay profitable even in a severe downturn. They are a near-duopoly in data-center HDD, each with ~44% revenue share, net cash, ~50% operating margins today and contracts out to 2028–31. **Seagate** leads in HAMR by ~1.5–2 years and is fully vertically integrated; **WD** spends more on R&D ($1.16B vs $0.76B), has the broadest mechatronics roadmap (14 platters, triple actuation, dual pivot, multi-head) and is the share leader in exabytes. **Toshiba** is investing to catch up but is ~2 years behind, depends on TDK for heads and is owned by private equity — a weaker 20-year bet.

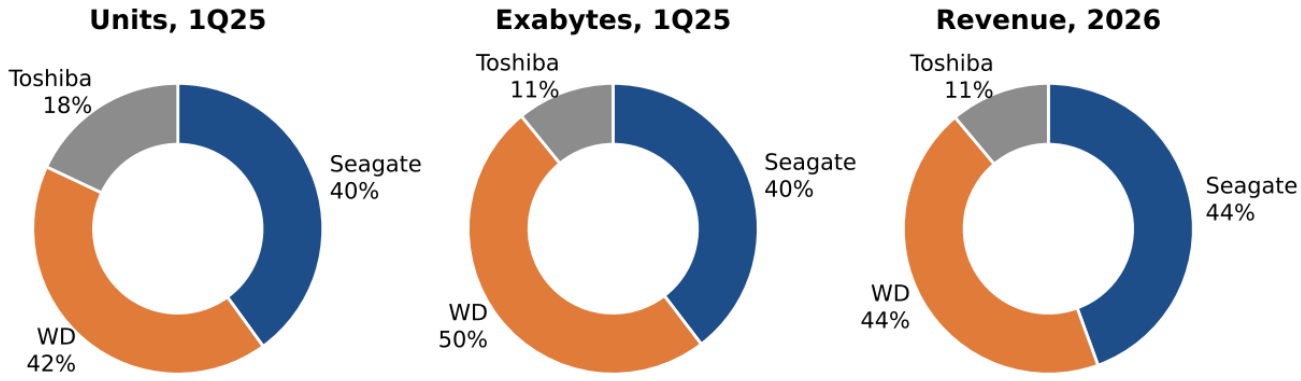
Scorecard

	Seagate (STX)	Western Digital (WDC)	Toshiba (HDD unit)
Latest qtr revenue (Sep-26)	$4.23B (+61%)	$4.38B (+55%)	~$1.1–1.4B [EST]
Gross / operating margin	57.8% / 50.7%	58.1% / 49.1%	Not disclosed (group target 10% op.)
FY2026 R&D	$755M (6% of rev.)	$1,161M (9% of rev.)	Not disclosed
Balance sheet	Debt ~$1.6B, net cash by end-2026	Net cash +$0.5B; investment grade	Private (JIP consortium)
Technology position	HAMR leader (44TB shipping)	ePMR/SMR leader; HAMR 1H27	MAMR; HAMR ~FY28; ~2 yrs behind
Vertical integration	Full (heads, media, lasers, SoC)	Heads + media in-house; own laser	Buys 100% of heads from TDK
Employees	~30,000 (83% Asia)	~40,000 (88% Asia)	HDD share of ~100k group
Survives severe downturn?	**Yes — clearly**	**Yes — clearly**	**Likely; less certain**
5-yr growth outlook	**Strong**	**Strong**	Moderate (share gains from low base)

Sources: company SEC filings and earnings calls (Oct 2026), TrendForce, Coughlin Associates, Blocks & Files. [EST] = analyst estimate. Not investment advice.

1. Industry structure

Consolidation finished a decade ago: from 9+ makers in 2000 to three. Seagate and WD split the data-center market almost evenly; Toshiba is a distant third in exabytes because it sells more low-capacity drives and is a generation behind in nearline.



Sources: TRENDFOCUS via StorageNewsletter (1Q25 units), derived exabyte share, Blocks & Files (Oct 2026). Rounded.

2. Seagate Technology (STX)

Ireland-domiciled, run from Fremont/Cupertino, California. Pure-play mass-capacity storage; no NAND fabs. Strategy: no new unit capacity — grow exabytes ~25%/yr through areal density (HAMR) with capex held at 4–6% of revenue.

Financial analysis

$M (GAAP)	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27E
Revenue	10,390	10,509	10,681	11,661	7,384	6,551	9,097	12,195	~18,500
Gross margin	28.2%	27.0%	27.3%	29.7%	18.3%	23.4%	35.2%	45.6%	~57%
R&D	991	973	903	941	797	654	724	755	~820
Operating margin	14.3%	12.4%	14.0%	16.8%	−4.6%	6.9%	20.8%	33.6%	~50%
Net income	2,012	1,004	1,314	1,649	−529	335	1,469	3,184	~8,000
Employees (k)	40.5	~40	~40	40.0	33.4	30.0	30.0	30.0	—

FY ends around 30 June. FY23 operating loss includes a $300M export-control penalty. FY27E = analyst estimate from Q1 actual ($4.23B) and Q2 guidance ($4.8B).

- **Mix:** Data Center ~81% of revenue ($3.4B in Sep-26 qtr); Edge-IoT ~19%. FY26 nearline exabytes 695 EB (+40%). Revenue per TB ~$19 (+~30% YoY).
- **Contracts:** most nearline exabytes allocated under LTAs through CY2028, some to 2030; CY2027 prices fixed. One customer = 14% of revenue.
- **Balance sheet:** debt cut from ~$5.0B to ~$1.6B; net leverage 0.2x; net cash expected by end-2026. Returns ~75% of FCF; dividend $0.76/qtr.
- **Opex is very lean:** ~$300M/qtr (~7% of revenue). R&D was cut 34% between FY19 and FY24 and has barely recovered — record profit, flat headcount.
- **Stock:** $668 (27 Oct 2026), ~$150B market cap; 52-week range $218–$1,145, so ~40% below the peak. ~17–18x run-rate EPS.

Technology analysis

- **Only maker shipping HAMR in volume.** Mozaic 3+ (30–36TB) qualified at all major clouds; Mozaic 4+ (44TB) shipping since March 2026 to the two largest; HAMR ~50% of nearline EB exiting 2026, target ~70% by mid-2027. Mozaic 5+ (50TB+) qualification late 2027.
- **Lab:** 6.9–8TB per platter demonstrated; target 10TB/platter → 100TB drives. Product steps ~4/5/6 TB per platter in 2027/28/29.
- **Vertical integration:** head wafer fabs (Northern Ireland, Minnesota), media (Singapore), in-house HAMR lasers/NFT and SoC; bought Intevac (media deposition tools) in 2025. Bloomberg reports interest in buying TDK's head business (not confirmed).
- **MACH.2** dual-actuator drives exist in small volume; relevance rises at 50TB+.
- **Risks:** HAMR scrap/yield is slowing cost-per-TB declines; lowest R&D intensity of the three.

3. Western Digital (WDC, branded "WD")

San Jose, California. Spun off its flash business as Sandisk in February 2025, making WD a pure-play HDD company (~88% of revenue from cloud). Built on the HGST (Hitachi/IBM) and legacy WD engineering organizations.

Financial analysis

$M	FY21	FY22	FY23	FY24	FY25	FY26	Q1 FY27	Q2 FY27 guide	FY27E
HDD revenue	8,216	9,040	6,255	6,316	9,520	12,919	4,381	5,100	~19,500
Gross margin (non-GAAP)	27.0%	29.4%	24.1%	29.8%	39.4%	49.1%	58.1%	60–61%	~59%
R&D (GAAP)	—	—	—	—	994	1,161	283	—	~1,200
Operating margin (non-GAAP)	—	—	—	—	24.4%	37.3%	49.1%	~52%	~51%
Free cash flow	—	—	—	—	1,432	3,511	1,377	—	~7,000
Employees (k)	—	65*	53*	51*	~40	~40			

FY21–FY24 = HDD segment (non-GAAP segment gross margin). *Headcount before FY25 includes flash. FY27E = analyst estimate.

- **Exabytes:** 248 EB in Sep-26 qtr (+22% YoY), largest in the industry; price per TB +28% YoY. Cloud $3.9B (88%).
- **Concentration:** top-3 customers = 44% of FY26 revenue (16/15/13%), top-10 = 73%. Top 7 customers have visibility to 2030–31.
- **Balance sheet transformed:** debt $7.4B (2024) → $0.79B; net cash ~$0.5B; Sandisk stake fully monetized; S&P and Fitch investment grade.
- **Targets already beaten:** Feb-2026 model (>50% GM, >40% op. margin, EPS >$20) is being met at run-rate. Buybacks $2.6B in FY26.
- **Stock:** ~$354 (22 Oct 2026), ~$132B market cap, ~15x forward run-rate EPS.
- **Layoffs during the boom:** 134 San Jose roles cut in Jan 2026 (post-split reshaping); restructuring charges ~$43M in Q1 FY27.

Technology analysis

- **Leader in conventional/SMR:** ePMR + OptiNAND + UltraSMR + 11 platters → 40TB UltraSMR drive in volume since mid-2026 (~50% of nearline shipments by early 2027). WD targets ~60% of exabytes on UltraSMR.
- **HAMR follower:** 44TB HAMR finishes qualification with 2 hyperscalers in Q1 2027, volume 1H27; 2nd generation (~5TB/platter) in 2H27. Roadmap 60TB (ePMR and HAMR, 14 platters) 2028 → 100TB HAMR 2029, using an in-house surface-emitting laser that saves z-height.
- **Broadest mechatronics roadmap:** High-Bandwidth Drive (multiple heads active at once, 2x now → 8x by 2030, sampling at 5 customers), Dual Pivot (two actuators, 2028), triple-stage actuation, power-optimized drive (−20% power), HDMR in the 2030s.
- **R&D intensity is the highest** of the three in dollars ($1.16B FY26). Key risk: 2027–28 HAMR execution versus Seagate.

4. Toshiba Electronic Devices & Storage (HDD)

Toshiba was taken private by the JIP consortium in December 2023. HDD results are not reported separately; group operating income rose from ¥39.9bn (FY2023) to ¥198.5bn (FY2024), partly on HDD.

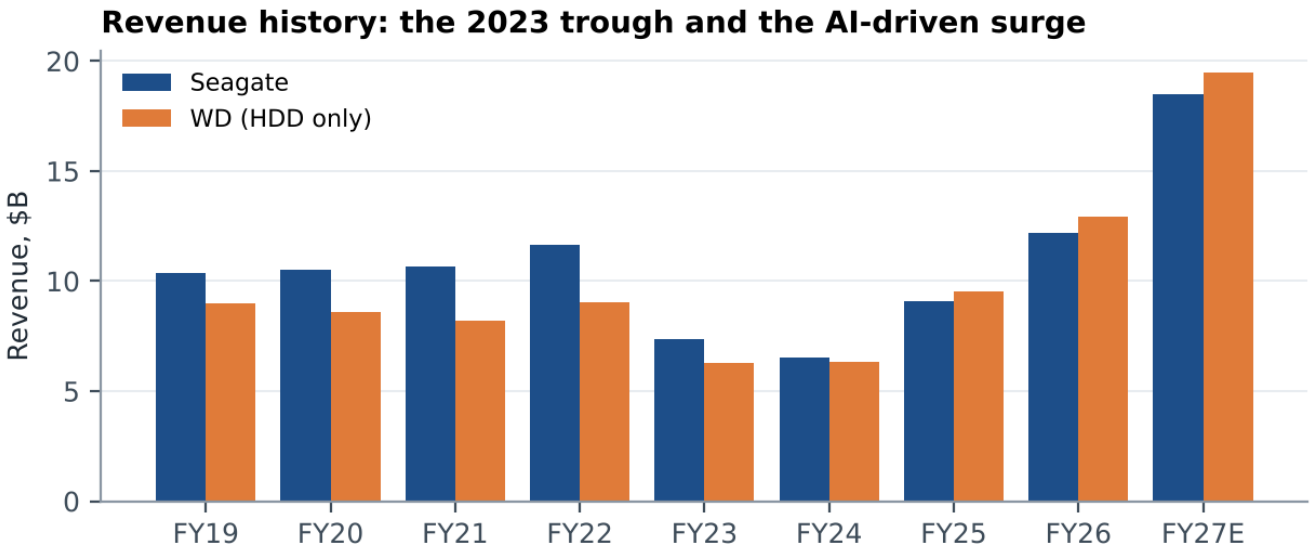
- **Share:** ~17–18% of units but only ~10–11% of exabytes and revenue. 1Q25: 5.2M drives, 41 EB, $945M. HDD revenue [EST] ~$3.8–4.2B in 2025, $4.5–5.5B in 2026.
- **Expanding, not exiting:** Oct-2026 announced ~¥60bn (~$380M) for Philippine plants to roughly double data-center output by FY2027; medium-term goal ~30% of industry production capacity. Seagate/WD shares fell 10–15% on the news.
- **Technology:** MAMR 24TB CMR / 28TB SMR shipping; 30–34TB SMR samples on 11 platters; 12-platter glass stack verified (Oct 2025). 40TB-class 2027, 45TB HAMR 2028, ~55–65TB around 2030. ~18–24 months behind Seagate.
- **Key dependency:** buys 100% of heads from TDK. Bloomberg (Oct 2026) reports Seagate and Toshiba competing to buy TDK's head business; if Seagate wins, Toshiba would source heads from its main rival.
- **People:** ~4,000 Japan job cuts in May 2024 (early retirement for staff 50+). Core HDD design in Yokohama/Kawasaki; a small San Jose team (self-servo-write, actuator system ID) — likely a few dozen HDD engineers outside Japan [EST].

5. Other HDD employers: the supply chain

Company	Role in HDD	Latest data	Relevance for a servo engineer
TDK / SAE Magnetics / Headway / Hutchinson	Only independent head maker; suspensions with microactuators	Segment sales ¥262.9bn (+17.6%), op. profit ~¥27bn (~8x); head unit may be sold	Low–medium: head/HGA, microactuator; mostly Asia
Resonac (ex-Showa Denko)	Largest merchant media supplier	Singapore plant 160M → 210M disks/yr for HAMR/MAMR	Low: materials/process
NHK Spring	Suspensions	FY3/26 sales ¥126.7bn, 20.6% margin; capex nearly triples	Low
Hoya	Glass substrates (100% of glass; ~40% of 3.5" DC)	~¥50bn Vietnam plant for FY2028	Low
Marvell / Broadcom	HDD SoC, read channel, preamps	Still supply all HDD makers; small HDD teams	Medium: servo demodulation, SoC — but wind-down risk

Real servo control-loop work stays inside Seagate, WD and Toshiba. Suppliers mostly hire materials and process engineers in Japan and Southeast Asia.

6. Growth potential, 2026–2031



WD FY19–20 approximate. FY27E = analyst estimate based on Q1 actual and Q2 guidance.

Base case [EST]	Seagate	WD	Toshiba HDD
Revenue 2026 (calendar)	~$15.8B	~$16.6B	~$4.5–5.5B
Revenue 2031	~$22–25B	~$23–26B	~$6–8B
CAGR 2026–31	~7–9%	~7–9%	~6–9% (share gains, lower price)
Operating margin 2031	~30–40% (from ~50%)	~30–40% (from ~49%)	Low double digits
Biggest swing factor	Keeping HAMR lead; 2028 LTA price reset	HAMR execution 2027–28; customer concentration	Head supply (TDK); catch-up on HAMR

Company forecasts apply Report 1's base case (industry ~$54B in 2031) with Seagate and WD holding ~42–45% share each and Toshiba rising toward ~13%. Margins are assumed to mean-revert as prices

normalize, but remain well above the 2018–19 peak (~15% operating margin) because opex is so lean.

7. Can they survive a whole-market downturn?

I stress-tested Seagate and WD against two scenarios from Report 1, starting from the FY27 run-rate. **Scenario A** (Report 1 bear case): revenue −35%, gross margin compresses to 35%. **Scenario B** (a repeat of 2023): revenue −45%, gross margin falls to 22%.

[EST]	Seagate — A	Seagate — B	WD — A	WD — B
Revenue	$12.0B	$10.2B	$12.7B	$10.7B
Gross profit	$4.2B	$2.2B	$4.4B	$2.4B
Opex (held flat)	$1.25B	$1.25B	$1.6B	$1.6B
Operating income	**~$2.9B (24%)**	**~$1.0B (10%)**	**~$2.8B (22%)**	**~$0.8B (7%)**
Balance sheet entering	Net cash	Net cash	Net cash	Net cash
Likely company response	Pause buybacks; hiring freeze	Restructuring 10–15%; plant underutilization	Pause buybacks; hiring freeze	Restructuring 10–15%; R&D trims

- **Survival is not in question for Seagate or WD.** Even a 2023-repeat leaves both profitable — in FY2023 they lost money because they carried $5–7B of debt, higher opex and a PC/consumer-heavy mix. Today both are net cash with opex at 7–9% of revenue.
- **Jobs are a different question.** In 2022–23 Seagate cut ~25% of headcount (40k → 30k) and executives took zero salary for six months; WD cut 65k → 53k. Both trimmed US R&D sites (Shakopee, Irvine) while Asian centers held. Expect a 10–15% restructuring in scenario B.
- **Toshiba** is private, so finances are opaque. It would likely survive inside the Toshiba group, but a downturn while it is spending to catch up — plus a private-equity owner looking for an exit — raises the risk of a sale or retreat of the HDD unit.
- **Suppliers** (TDK heads, Marvell/Broadcom HDD chips) are the most exposed: HDD is a non-core unit for their parents, so a downturn could accelerate divestiture or wind-down.

Long-term ranking (10–20 years) on business strength: 1. Seagate and **1. WD** (tie — Seagate stronger in HAMR and integration, WD stronger in R&D spend and roadmap breadth), **3. Toshiba**, then supply-chain firms. If the industry ever consolidates to two, the two will almost certainly be Seagate and WD.

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